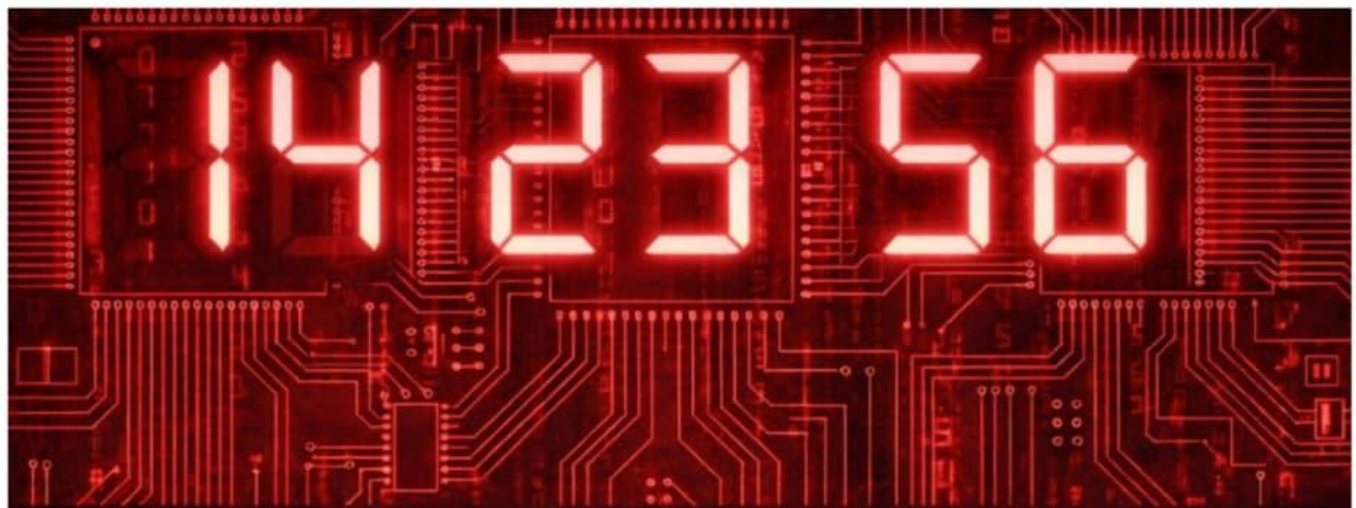




DIGITAL CLOCK PROJECT

Electronics and Logic Gates

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1) Objectives

The main objectives of the project is :

- To assemble a **working digital clock** using basic logic integrated circuits.
- To understand how counters and display drivers work together to create a time display.
- To learn how TTL logic ICs like the **74LS90** and **74LS47** can generate and drive digits.
- To build a circuit that displays seconds and minutes on 7-segment displays.

2) Introduction

In this project, shows **step-by-step how to make a practical digital clock** using discrete integrated circuits rather than a microcontroller board. The design uses classic TTL components — counters and display drivers — to increment time and display it on 7-segment displays.

A digital counting circuit is designed to demonstrate the basic working principle of a digital clock. The **555 Timer IC** is used as a clock pulse generator, producing periodic square-wave signals. These pulses are then fed into a **BCD Decade Counter (74LS90)**, which counts from 0 to 9 repeatedly.

The output of the counter is in binary-coded decimal (BCD) format, which cannot be directly understood by humans. Therefore, a **7447 BCD to Seven-Segment Decoder** is used to convert the BCD signals into appropriate signals for the **seven-segment display**, allowing decimal numbers to be visually displayed.

This project was simulated using Proteus software and can later be implemented physically on a breadboard for real-time testing.

This type of project is great for beginners who want to explore **hardware logic**, learn how digital counters work, and gain hands-on experience with breadboards and circuit wiring.

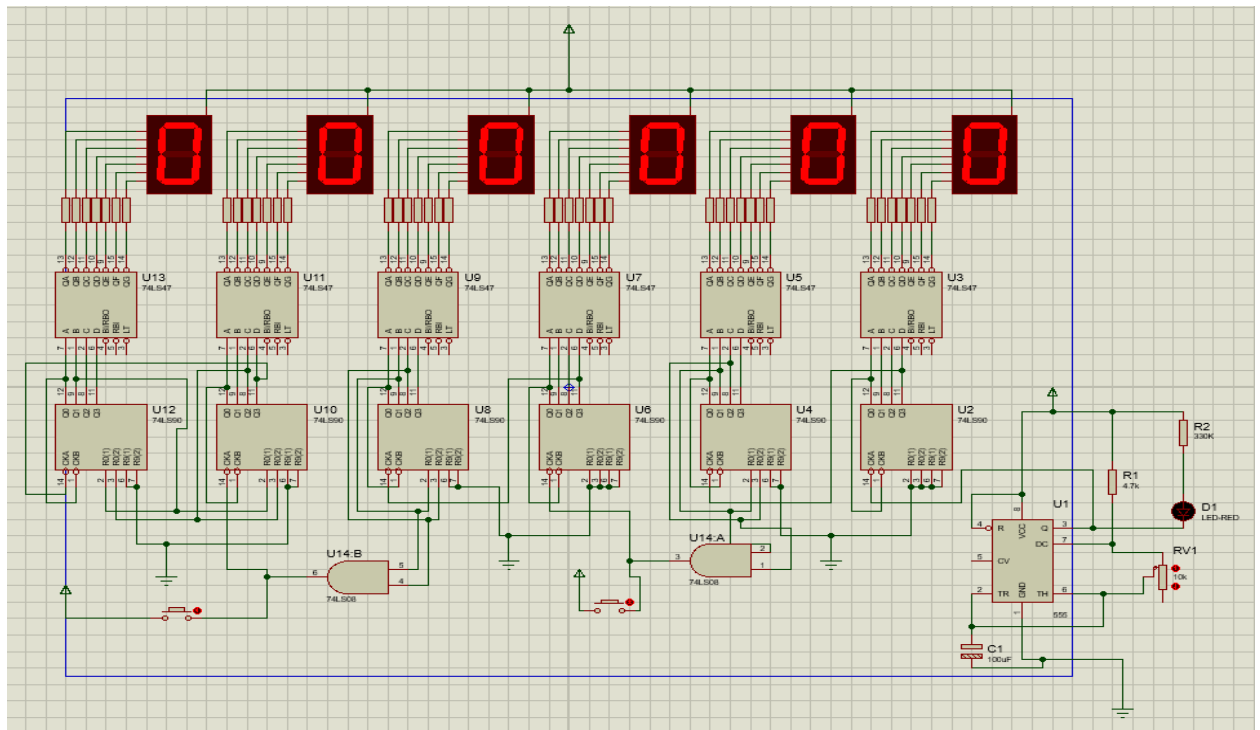
3) Chapters

□ Chapter 1: Overview of Digital Clock Concept

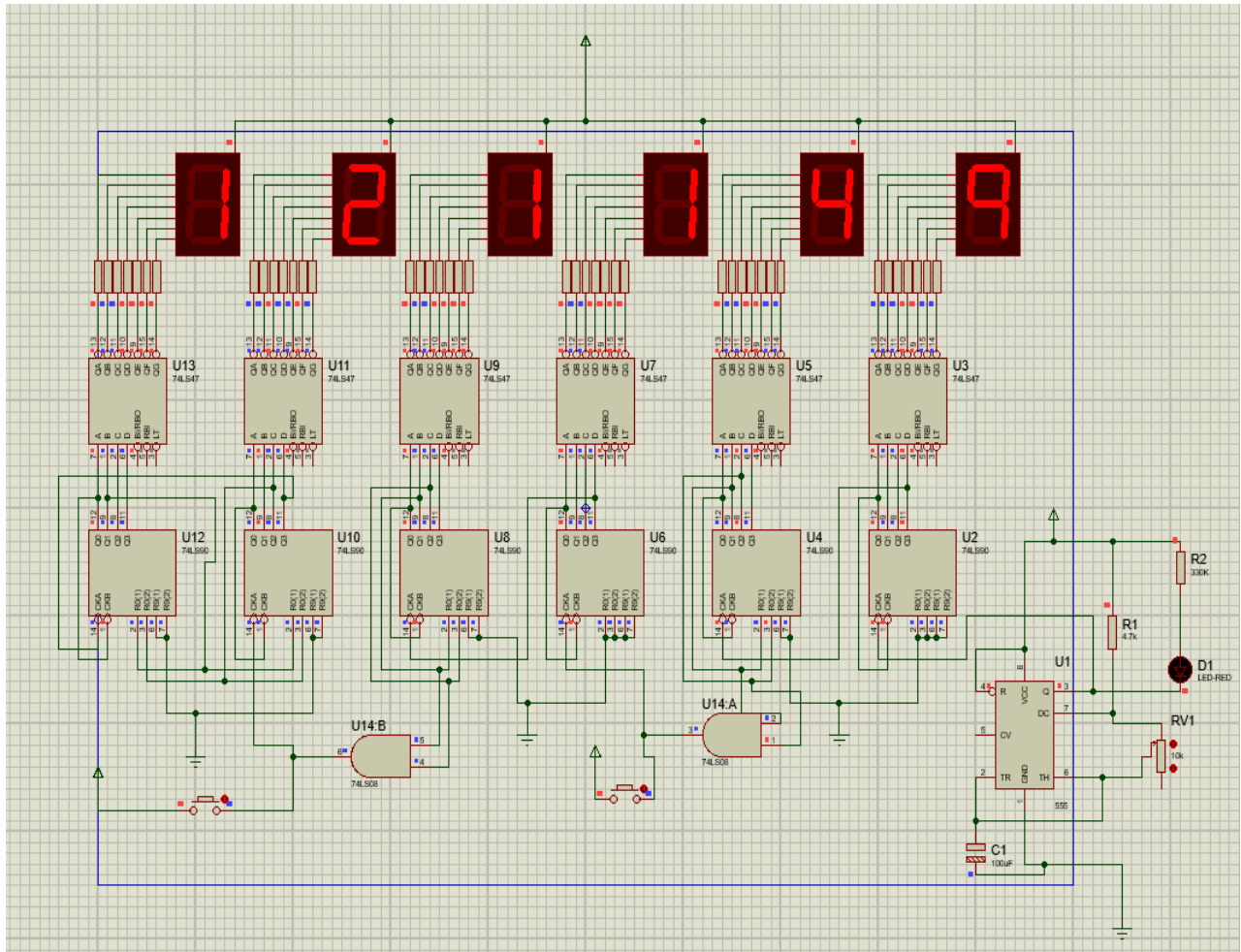
The basic concept of a digital clock circuit:

- **Counters:** ICs like the 74LS90 are used to count pulses from a clock source to represent seconds and minutes.
- **Driver:** The 74LS47 decodes the counter outputs into the proper signals to light up 7-segment displays.
- **Display:** 7-segment displays show the digits for seconds and minutes.
- **Clock Source:** A simple oscillator (using a 555 timer) produces pulses to drive the counters.
- **The counters increment as follows:** 1 Hz pulse → Seconds Units → Seconds Tens → Minutes Units → Minutes Tens → Hours Units → Hours Tens

Schematic Diagram Image



Simulation Image



The simulation was produced using circuit design software (**Proteus**).

- Correct counting from 00 to 59 seconds
- Correct rollover of minutes
- Hours display switching after 60 minutes
- Reset logic working at the correct state

□ Chapter 2: Components

□ Components List:

- 6 × **74LS90** decade counters
- 6 × **74LS47** BCD-to-7-segment decoders
- 1 × **NE555 Timer IC**
- 1 × **74LS08 AND Gate**
- 6 × Common-anode 7-segment displays
- Resistors:
 - 470 Ω ×6
 - 4.7 k Ω ×1
 - 330 Ω ×1
- Capacitors:
 - 100 μ F
 - 680 μ F
- 2 × pushbuttons
- 5 V power supply
- Connecting cables and breadboard

★ Function of Each Component

- **74LS90:** Counter IC that receives clock pulses and increments a BCD count.
 - **74LS47:** Takes BCD and drives a 7-segment display.
 - **NE555:** Probably used to generate a stable pulse for seconds.
 - **74LS08:** AND logic gate for combining or gating signals.
 - **Pushbuttons:** Likely used for setting or resetting time.
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Chapter 3: Wiring

This chapter explains how components are interconnected:

- **Power and Ground:** Common 5 V supply to all TTL ICs (common Vcc and GND).
- **Clock Source:** The NE555 output connects into the first counter to produce a one-pulse-per-second clock.
- **Counters:** Cascaded 74LS90 ICs count up to 60 (seconds and minutes).
- **Display Drivers:** Each counter's BCD outputs connect to the input of a 74LS47.
- **7-segment Displays:** Outputs from the 74LS47 go to the segments of each display.
- **Buttons:** For reset/adjust functions.

CHAPTER 4: TESTING

Testing Procedure:

1. Power was applied (5V)
2. The LED connected to the 555 output showed blinking at 1 Hz
3. Seconds were observed: From 00 to 59
4. Minutes were observed: After 60 seconds, minutes increment correctly
5. Hours were tested to make sure: Rollover worked correctly depending on chosen mode (12 or 24 hour)
6. A reset button was tested to bring values back to 00:00:00

Result: The clock counted correctly through full second and minute cycles.

5) Summary

In summary, this report demonstrates how to build a **digital clock circuit without a microcontroller**, using classic TTL ICs. It covers:

- A list of components and their functions.
- An explanation of the counting mechanism and display interface.
- Wiring and breadboarding of a real-world clock.
- A functional test to ensure correct time counting.

This project is excellent for electronics students and hobbyists wanting to understand digital logic, timing circuits, and hardware implementation of timekeeping. The Report provides enough guidance to build and test your own clock while allowing space for you to innovate or simulate the design.
